

Table 1: Parameter-level sparsity is a BF16 precision artifact
BF16 rounds small weight updates to zero — the effect is identical across RL and SFT

Optimizer	Objective	% Changed (FP32)	% Changed (BF16)	% Changed (Mixed)	Rank@90% (FP32)	Rank@90% (BF16)	Rank@90% (Mixed)
AdamW	GRPO	100.0%	2.1%	100.0%	190	732	329
Muon	GRPO	99.9%	0.5%	99.9%	752	590	748
AdamW	SFT	100.0%	2.1%	100.0%	450	717	379
Muon	SFT	99.9%	0.4%	99.9%	794	492	730